Week 1
Introduction

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Introduction to Phonetics and Phonology

The production of speech sounds

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Articulators

- **The larynx (the voice box / the Adam's apple)**
  - a tube-shaped organ in the neck that contains the vocal cords. It's located between the pharynx and the trachea.
- **The vocal cords**
  - Two small bands of elastic tissue, which can be thought of as two flat strips or rubber, lying opposite each other across the air passage in the larynx.
  - The inner edges of the vocal cords can be moved towards each other so that they completely cover the top of the wind pipe (the trachea), or can be drawn apart so that there is a gap between them.

Articulators

- **The pharynx**
  - The space behind the tongue, immediately above the larynx, reaching up towards the nasal cavity.
- **The palate**
  - Forms the roof of the mouth.
  - Separates the mouth/oral cavity from the nose/nasal cavity.
  - The front part is hard, while the back part is soft.
  - The soft palate (velum) can be raised so that it makes a firm contact with the back wall of the pharynx.

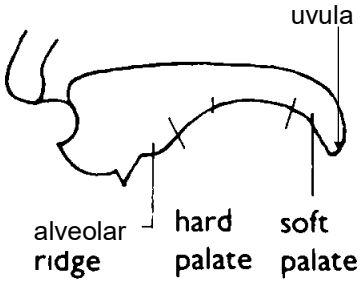


Fig. 5 The parts of the palate

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Articulators

- **The teeth**
  - The lower front teeth are not very important in speech.
  - The upper front teeth are more frequently used in English.
- **The tongue**
  - The most important of the speech organs because it has the greatest variety of movement.
  - Divided into five parts: tip, blade, front, back (and root).

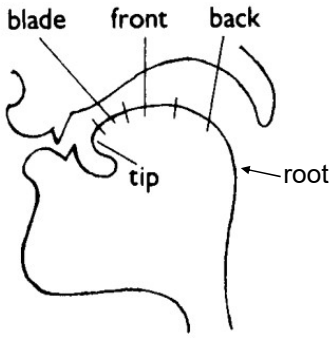


Fig. 6 The parts of the tongue

Articulators

- **The lips:**
  - Consist of the upper lip and lower lip.
  - Can take various different positions:
    - brought firmly together so that they completely block the mouth.
    - the lower lip can be drawn inwards to touch the upper front teeth.
    - kept apart either flat or with different amount of rounding.
    - pushed forward to a greater or lesser extent.

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Articulators

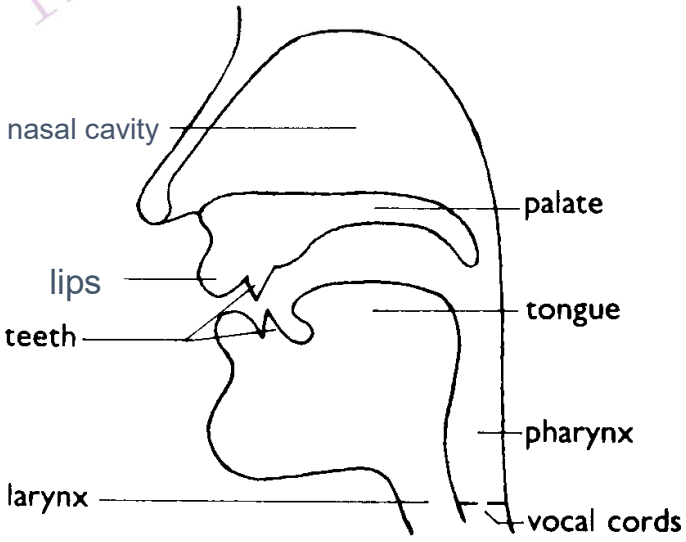
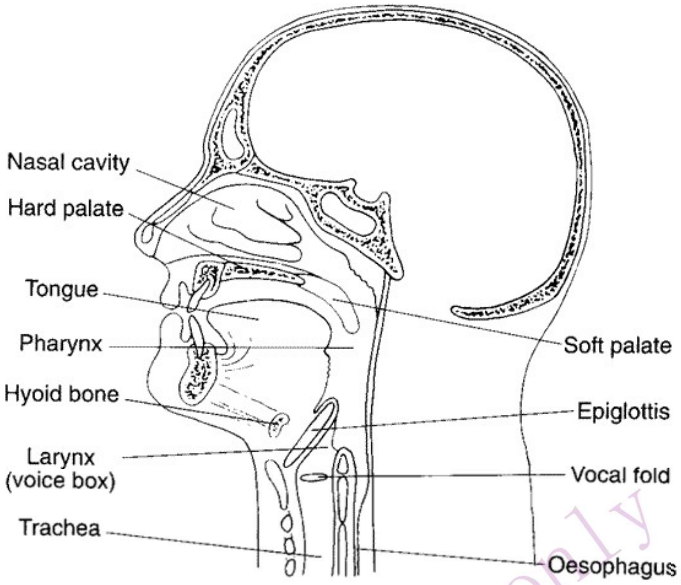


Fig. 1 The speech organs

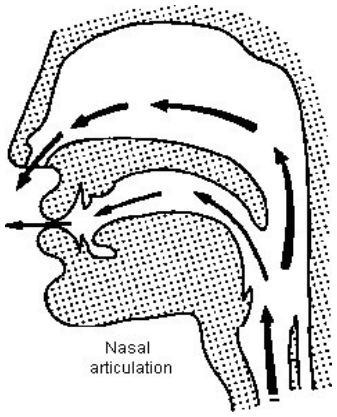
Articulators



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How are speech sounds produced?

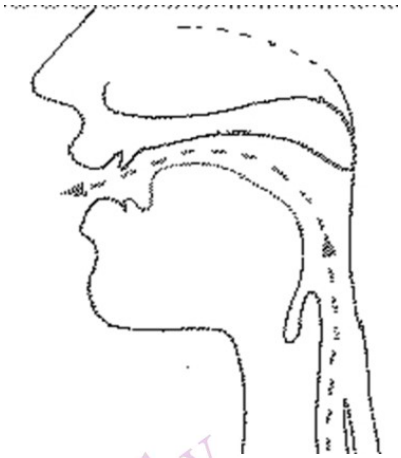
- When we are making sounds, the air from the lungs comes up through the windpipe/trachea and arrives at the larynx.
- Then it goes through the vocal cords into the pharynx and up to the uvula.
- At this point, the air may go in either way:
  - It may go into the oral cavity & get out through the mouth.
  - Or it may go into the nasal cavity & get out though the nose.



Oral vs. Nasal sounds

1. **How are oral sounds produced?**

- Oral sounds are the sounds in the production of which the soft palate is raised, blocking off the nasal cavity so that ***the airstream can only get out through the mouth***.
- E.g. /b, g, ʊ, ʃ, æ/

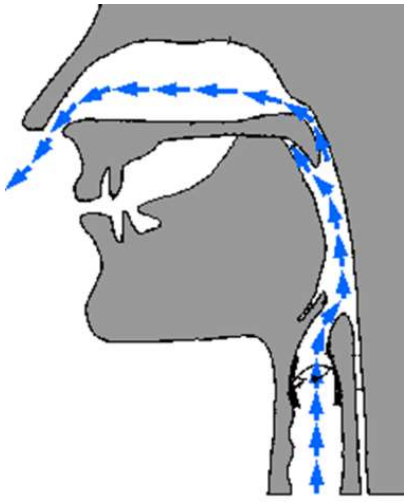


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Oral vs. Nasal sounds

2. **How are nasal sounds produced?**

- Nasal sounds are the sounds in the production of which the soft palate is lowered, blocking off the oral cavity so that ***the airstream can only get out through the nose.***
- There are only three nasal sounds in English: /m, n, ŋ/



Consonants vs. Vowels

3. **How are consonant sounds produced?**

   When we are making sounds, if two articulators come together, obstructing the air-stream and the air-stream cannot get out freely, we have consonant sounds. In other words, there is some stricture or closure of the air stream.

   E.g. /k/, /f/, /b/, /m/

4. **How are vowel sounds produced?**

   When we are making sounds, if there is no obstruction to the flow of air as it passes from the larynx to the lips, and the air can get out freely, then we have vowel sounds. In other words, they are produced with open articulation.

   E.g. /ɒ/, /ɜː/, /æ/, /ɪ/

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Consonants vs. Vowels

- Problem in English: some consonants do not really obstruct the flow of air more than vowels do.

  Eg. the beginning sound of *'hay'* and *'way'*

=> The classification of sounds can also be based on their distribution in the syllable.

Voiced sounds vs. Voiceless sounds

5. **How are voiced sounds produced?**
   When we are producing sounds, the air-stream goes through the vocal cords. If the vocal cords come together, obstructing the air-stream, the air-stream cannot get out through them freely and it makes them vibrate, then we have voiced sounds.
   E.g. /d/, /v/, /m/
6. **How are voiceless sounds produced?**
   When we are making sounds, the air-stream goes through the vocal cords. If the vocal cords come apart, they are open. The air-stream can go out through them freely and it does not make them vibrate, then we have voiceless sounds. E.g./s/, /t/, /ʃ/

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